SPECIFICATION

Cross-Reference to Related Applications

This application is related to the inventions described in the U.S. Patent Applications for "A Central Processing Unit Architecture for Native Parametric and Symbolic Computation," "A Memory System for the Storage and Retrieval of Parametric and Symbolic Data Representations," and "A Method and System for High-Radix Physical Layer Data Encoding and Representation," the contents of which are hereby incorporated by reference in their entirety. The present invention utilizes the principles of said applications to create a complete networking paradigm.

Background of the Invention

Conventional networking protocols, such as the TCP/IP suite, are fundamentally designed to transport streams of binary data. All information, regardless of its complexity or required precision, must first be serialized into a sequence of bits. This bit-stream is then segmented into packets, transmitted across a network, and reassembled at the destination. While robust, this model forces the premature evaluation and approximation of data. A high-precision number, for instance, must be converted to a standard floating-point representation before it can even be placed in a TCP packet, losing its perfect fidelity at the source. The network's sole function is to transport these already-approximated bits.

Summary of the Invention

The present invention provides a complete networking protocol stack and system designed to natively transport parametric and symbolic data. It redefines the concept of a data packet from a container of bits to a container for an unevaluated mathematical "recipe."

The system defines a new packet structure, the "parametric packet." This packet contains a header and a payload. The header may contain routing and control information in a symbolic or high-radix format. The payload is a variable-length field designed to hold a data structure representing a mathematical expression, such as '7/16' or a formula for calculating a value.

Network infrastructure, such as routers and switches, are equipped with a parametric forwarding engine. This engine reads the symbolic header of an incoming parametric packet and determines the appropriate outbound port based on its routing table, which is also indexed symbolically. Critically, the forwarding engine does not parse, evaluate, or modify the packet's payload. The unevaluated "recipe" passes through the node untouched.

This method ensures end-to-end precision. Data generated by a parametric CPU can be packetized, transmitted across a multi-hop network, and received by another parametric system without any loss of fidelity. The evaluation to a numerical value, if needed, occurs only after the data has been successfully delivered to the target application.

Brief Description of the Drawings

FIG. 1 is a block diagram comparing a traditional network data flow with the parametric network data flow of the present invention.

FIG. 2 is a conceptual diagram of a parametric packet, illustrating its symbolic header and unevaluated payload.

Detailed Description of the Invention

Referring to **FIG. 1**, the fundamental difference in data flow is illustrated. In a traditional network (100), a high-precision number (110) is first approximated into a fixed-width binary format (115). This lossy data is placed in a standard IP packet, traverses a router (120), and arrives at the destination with its precision already compromised. In the parametric network (150), the same high-precision number is encapsulated as a symbolic "recipe" (160) into a parametric packet. The packet traverses a parametric router (170) which forwards it based on its header without inspecting the payload. The packet arrives at the destination with its perfect, unevaluated "recipe" (160) fully intact.

FIG. 2 shows the structure of a parametric packet (200). The packet consists of a header (210) and a payload (220). The header contains fields such as 'Symbolic Destination Address' (212) and 'Symbolic Source Address' (214). These fields contain high-radix or symbolic representations, not traditional IP addresses. The payload (220) is a variable-length container holding the unevaluated data structure (230), for example, the "recipe" for a number. The packet transports the concept of the data, not its evaluated value.